

TaxonVision-MLOps: Architectural Concept Design End-to-End Species Identification Pipeline

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Abstract

This document specifies the architectural concept design for **TaxonVision-MLOps**, an automated, production-grade Machine Learning Operations platform for fine-grained biological species identification from citizen-science imagery (e.g., iNaturalist, GBIF). Grounded in the principle that a simple, versioned, monitored, deployed, and automatically retrainable pipeline is superior to an isolated notebook model, this design addresses four graduated operational levels: closed-chain ingestion under strict Creative Commons licensing, Pareto-optimal backbone benchmarking and zero-allocation ONNX Runtime inference, closed-loop telemetry and taxonomic mutation reconciliation, and rigorous statistical safety guarantees via distribution-free Split Conformal Prediction ($1 - \alpha$ coverage) combined with active learning triage.

1 Introduction and Problem Domain

Ecological citizen science platforms such as iNaturalist record hundreds of millions of biodiversity observations [5]. Developing an automated identification service for this domain presents fundamental challenges:

- **Long-Tailed Taxonomic Distributions:** Taxa frequencies adhere to extreme power-law distributions where thousands of rare species possess few reference images.
- **Licensing and Provenance Governance:** Only observations distributed under open licenses (CC0, CC-BY, CC-BY-NC) may be ingested. Static domain assets lacking open licenses must be rejected, and attribution metadata must be preserved throughout the pipeline.
- **High-Stakes Decision Uncertainty:** Raw softmax probabilities are notoriously overconfident and uncalibrated. Mission-critical deployment demands mathematically guaranteed confidence sets with automatic referral to human experts for ambiguous observations.

2 Graduated Architectural Levels

2.1 Level 1: Restricted Perimeter and Closed Chain

The initial baseline establishes an automated, end-to-end chain over a representative subset of well-documented taxa:

- **Data Ingestion:** Streaming and caching open-access observation metadata and images from the iNaturalist Open Data AWS S3 bucket.

- **Feature Extraction:** Fixed pre-trained feature extractors (e.g., lightweight convolutional or transformer architectures via `timm`, BioCLIP [4]) with a modular classification head.
- **Licensing Enforcement:** Automated cryptographic validation ensuring no non-open license images enter the dataset artifact.
- **Unified Web Service:** Minimal FastAPI serving endpoint exposing REST predictions alongside an HTMX operator dashboard.

2.2 Level 2: Scalability, Cost, and Pareto Optimization

Scaling to hundreds or thousands of taxa introduces cost and latency trade-offs:

- **Pareto Backbone Benchmarking:** Systematic evaluation of candidate backbones (DINOv3, BioCLIP-2, DINOv2, MobileNetV4) balancing Top-1 accuracy against $p95$ latency and computational cost, trained with Class-Balanced Loss [2].
- **Inference Optimization:** Exporting models to ONNX Runtime with dynamic INT8 and FP16 quantization, enforcing zero-allocation hot paths.
- **Large Dataset Versioning:** DVC-managed content-addressable storage maintaining reproducible dataset snapshots.

2.3 Level 3: Feedback Loops and Production Observability

Continuous operations require robust monitoring and label reconciliation:

- **Closed Feedback Loop:** Automated ingestion of community-validated identifications over time to measure real-world performance degradation.
- **Taxonomic Mutation Handling:** Linnaean taxonomy revisions (splits, lumps, reclassifications) are tracked and mapped via DarwinCore taxon identifiers.
- **Telemetry:** Exporting Prometheus metrics for request latency, prediction entropy, human referral ratios, and feature embedding drift.

2.4 Level 4: Mathematical Robustness, Cold Start, and Active Learning

The safety layer protects downstream users and prioritizes human expert effort:

- **Out-of-Distribution (OOD) Detection:** Energy-based scoring filters photographs devoid of organisms, corrupted images, or uncatalogued taxa.
- **Few-Shot & Cold Start Protocol:** Embedding nearest-neighbor matching for rare taxa with limited photographic records.
- **Active Learning Triage:** Hybrid query strategies (BADGE and CoreSet) [3] rank ambiguous samples to maximize information gain per human annotation.
- **Low-Latency Explanations:** Grad-CAM heatmap generation constrained to sub-25ms response budgets.

3 Mathematical Invariants: Conformal Prediction

Rather than relying on raw softmax scores $\hat{f}(X)$, we apply Split Conformal Prediction [1] to construct prediction sets $\hat{C}(X_{n+1}) \subseteq \mathcal{Y}$ satisfying:

$$P(Y_{n+1} \in \hat{C}(X_{n+1})) \geq 1 - \alpha \quad (1)$$

for a user-specified significance level $\alpha \in (0, 1)$.

Given n holdout calibration samples $(X_i, Y_i)_{i=1}^n$, we define the non-conformity score:

$$s_i = 1 - \hat{f}(X_i)_{Y_i} \quad (2)$$

The critical calibration threshold $\hat{q}$ is computed as the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of $\{s_1, \dots, s_n\}$. For a test image X_{test} , the output prediction set is:

$$\hat{C}(X_{\text{test}}) = \left\{ y \in \mathcal{Y} : 1 - \hat{f}(X_{\text{test}})_y \leq \hat{q} \right\} \quad (3)$$

If $|\hat{C}(X_{\text{test}})| = 0$ (indicating an atypical or OOD observation) or $|\hat{C}(X_{\text{test}})| > k_{\text{max}}$ (indicating high taxonomic ambiguity), the observation is automatically referred to human triage.

4 Conclusion

The TaxonVision-MLOps concept design bridges cutting-edge representation learning and rigorous statistical guarantees within an automated, reproducible MLOps lifecycle.

References

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